

TRACE-TAR: Reusable Video Evidence and Shared Event Memory for Transportation Anomaly Reasoning

Abhineet Kumar Pandey¹ , Ming-Ching Chang¹ , and Jun-Wei Hsieh² 

¹ Department of Computer Science,
University at Albany, State University of New York,
Albany, NY 12222, USA
{apandey,mchang2}@albany.edu

² National Yang Ming Chiao Tung University,
Daxue Rd, Guangming Village, East District,
Hsinchu City, 30010, Taiwan
jwhsieh@nycu.edu.tw

Abstract. Transportation anomaly reasoning requires a model to identify unusual events, localize them in time, and answer heterogeneous questions about participants, causes, consequences, and scene context. We present **TRACE-TAR**, a multimodal adaptation pipeline that decodes each video once, ranks coarse temporal windows using label-free motion, appearance-change, and interaction cues, and exposes automatically mined tracks and relations as uncertain shared context. We adapt Qwen3-VL-8B with low-rank adaptation (LoRA) and answer-token supervision on a video-disjoint local split of the AI City Challenge 2026 Traffic Anomaly Reasoning dataset. After media validation, the split contains 39,720 training and 4,296 validation records. Under identical evidence, prompting, formatting, and decoding, adaptation improves the nine-task local mean from 0.2922 to 0.5587; binary and multiple-choice accuracy reach 94.4% and 88.3%, respectively, and temporal localization reaches 0.279 mean temporal IoU. Video-clustered bootstrap intervals confirm the categorical gains. The comparison establishes the value of task adaptation under the executed evidence protocol.

Keywords: Transportation anomaly reasoning · Video question answering · Evidence grounding · Cross-task consistency

1 Introduction

Transportation surveillance videos present a demanding setting for multimodal reasoning. An anomalous event may be brief, spatially localized, partially occluded, or visible only through subtle changes in the motion and interactions of road users. Understanding such events requires more than assigning an anomaly score: a model may also need to identify the participants, determine when the event occurred, explain its cause and consequences, and summarize the

scene coherently. The AI City Challenge 2026 Traffic Anomaly Reasoning benchmark formalizes this setting through ten task types spanning binary classification, multiple-choice reasoning, open-ended question answering, causal analysis, temporal localization and description, scene description, and video summarization [1].

Two challenges are particularly important in this setting. First, uniform frame sampling may omit short or visually subtle events, especially when a long video is represented by only a small number of frames. This limitation is consequential for questions that depend on motion transitions, interactions, or the ordering of pre-event and post-event states. Second, questions associated with the same video are typically processed and evaluated independently, even though they describe different aspects of a common underlying event. As a result, a model may produce individually plausible but mutually incompatible predictions, such as denying that an anomaly occurred while describing the same event in a causal explanation or summary. Independent question processing also makes it difficult to determine whether the model’s answers are supported by the relevant visual evidence or primarily reflect linguistic priors and dataset regularities.

Recent work has advanced several related directions. Video anomaly reasoning extends anomaly scoring toward explicit event descriptions and causal explanations [4]; evidence-aware VideoQA evaluates whether an answer is supported by the correct temporal or spatio-temporal evidence [5, 7]; and graph-based VideoQA represents entities and relations at multiple temporal scales [8]. TRACE-TAR does not replace these grounding or graph-reasoning formulations. Its narrower focus is the interface between them in a multi-task traffic benchmark: a single label-free index and uncertain relation summary are reused for all questions about a video, while task-specific prompts determine the required output. This distinction matters in surveillance footage, where small objects, occlusion, and fragmented motion tracks make automatically mined relations unreliable.

We present **TRACE-TAR**, a reproducible multimodal pipeline for transportation anomaly reasoning. TRACE-TAR decodes each video once to construct a reusable evidence index containing timestamps, motion cues, appearance changes, motion-derived regions, tracklets, and candidate interactions. For each annotation record, the system selects temporally diverse evidence using video-derived saliency signals and constructs an agent-centric event memory that summarizes candidate participants, relations, temporal extent, and confidence estimates. Because these quantities are obtained automatically and may be noisy, the serialized memory is explicitly presented to the model as uncertain supporting context rather than as ground truth. The same video-level memory is reused across the heterogeneous questions associated with a video, providing a common representation of the event while preserving task-specific prompts and output formats.

We adapt Qwen3-VL-8B using low-rank adaptation and answer-token supervised fine-tuning. Training and validation are separated at the video level to prevent annotations from the same video from appearing in both partitions.

After media verification, the resulting local experiment contains 39,720 training records and 4,296 validation records. Under matched evidence-selection, prompting, formatting, and decoding settings, the adapted model improves the nine-task local benchmark mean from 0.2922 for the frozen backbone to 0.5587. It achieves 94.4% binary-question accuracy and 88.3% multiple-choice accuracy, while temporal localization reaches a mean temporal intersection over union of 0.279. Temporal localization is reported separately because it is not included in the nine-task benchmark mean.

The main contributions of this work are as follows:

- We develop a reusable video-evidence indexing pipeline that combines motion, appearance-novelty, and interaction cues to support deterministic and temporally diverse evidence selection.
- We introduce an uncertain agent-centric event memory that shares candidate participants, relations, temporal structure, and confidence information across transportation-reasoning tasks.
- We report a leakage-resistant, video-disjoint evaluation of frozen and LoRA-adapted Qwen3-VL under identical inputs and decoding, together with video-clustered uncertainty estimates and cross-task failure analysis.

2 Related Work

Video anomaly detection and reasoning. Classical video anomaly detection seeks to identify events that deviate from learned patterns of normal behavior, often under weak or video-level supervision. A central challenge is anomaly locality: an unusual event may occupy only a small spatial region and a short temporal interval within an otherwise ordinary video [6]. Recent work has expanded this problem from anomaly scoring to explicit multimodal reasoning. In particular, VAD-R1 introduces a perception-to-cognition reasoning framework that enables post-trained vision–language models to produce event-level explanations rather than only anomaly scores [4]. TRACE-TAR considers a related but distinct setting in which multiple questions about the same video describe complementary aspects of a common event. Accordingly, the proposed pipeline constructs a shared video-level representation that can be reused across anomaly classification, temporal localization, causal reasoning, scene description, and summarization.

Temporal grounding and evidence-aware VideoQA. Temporal grounding aims to identify the video interval that corresponds to a natural-language query [9]. This problem is closely related to long-form VideoQA, where uniform frame sampling may omit short or visually subtle events. Modern approaches therefore employ adaptive frame or clip selection to prioritize temporally relevant evidence. EG-VQA jointly evaluates semantic answer quality and temporal evidence localization, demonstrating that a model may generate a correct answer while attending to an incorrect segment of the video [5]. Evidence-Backed VQA

extends this perspective by associating answers with tracked spatio-temporal evidence masks [7]. In contrast, TRACE-TAR uses coarse temporal windows and exposes track relations as uncertain text shared across questions; it does not claim fine-grained spatio-temporal grounding.

Structured representations for video reasoning. Structured video representations make objects, actions, trajectories, and relations explicit, thereby complementing global visual embeddings that may obscure fine-grained interactions. Graph-based formulations have been effective for multi-granular VideoQA because they support reasoning over entities and relations at different temporal scales [8]. Their application to transportation surveillance is nevertheless challenging: road users may be small, partially occluded, or blurred; detections may be unreliable; and long trajectories may fragment across frames. TRACE-TAR therefore represents agents and candidate interactions as uncertain observations with associated confidence estimates, rather than treating automatically constructed tracks and relations as ground truth. The resulting event memory serves as auxiliary context for multimodal reasoning while preserving access to the selected visual evidence.

Parameter-efficient multimodal adaptation. Large vision-language models provide a flexible foundation for reasoning over interleaved images, video, and text. Qwen3-VL supports multimodal inputs and temporal alignment mechanisms suitable for video understanding tasks [2]. However, full-parameter fine-tuning of such models is computationally expensive. Low-rank adaptation introduces trainable low-rank updates to selected network projections while keeping the pretrained backbone frozen, substantially reducing the number of optimized parameters [3]. We use LoRA to adapt the 8B Qwen3-VL backbone to the heterogeneous output formats of transportation anomaly reasoning. In this work, parameter-efficient adaptation enables reproducible experimentation, while the methodological focus is the organization of reusable temporal evidence and shared event-level context.

3 Problem Formulation

Let a video be $V = \{I_t\}_{t=1}^T$ and let $\mathcal{Q}_V = \{(q_i, \tau_i, y_i)\}_{i=1}^{N_V}$ denote its questions, where $\tau_i \in \mathcal{T}$ is a task type and y_i is the reference answer. The task set contains binary and multiple-choice questions and eight open or structured reasoning tasks, including localization, causal linkage, temporal description, scene description, and summarization. A model F predicts

$$\hat{y}_i = F(q_i, \tau_i, \widetilde{\mathcal{M}}_V, \mathcal{E}_i), \quad (1)$$

where $\widetilde{\mathcal{M}}_V$ denotes the model-visible subset of event memory and $\mathcal{E}_i$ contains at most four selected frames. We evaluate each task with its benchmark metric and audit cross-task errors separately.

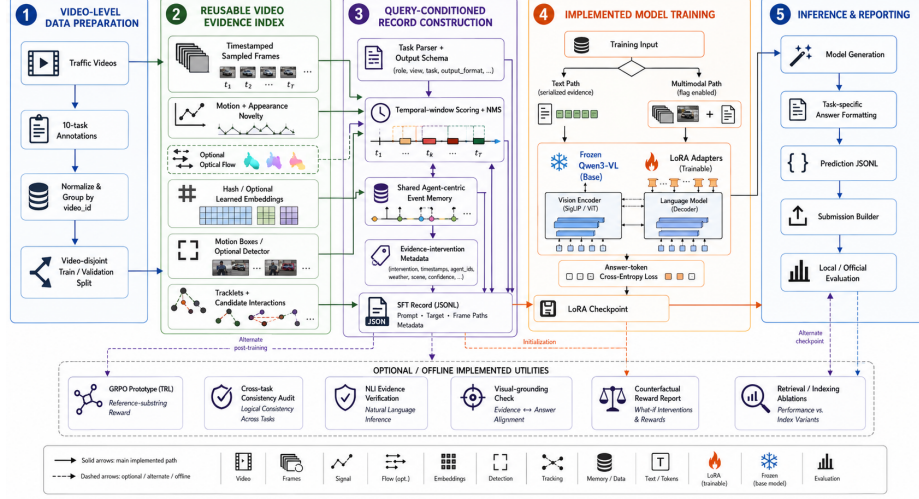


Fig. 1: Overview of the executed TRACE-TAR pipeline. A label-free video index supplies saliency rankings, motion tracks, and candidate relations. The model receives up to four selected frames and an uncertain textual relation summary shared by all questions for that video. Ground-truth answers are used only as supervised targets and evaluation references.

4 TRACE-TAR

4.1 Framework Overview

Figure 1 summarizes the executed training and inference path. Annotations are normalized and partitioned by `video-id`; videos are processed independently of those annotations to construct a reusable index. For each question, the index supplies ranked temporal windows and a common textual summary of tracks and candidate interactions. The question, selected frames, and shared summary form the model input, while the annotation answer is used only as the supervised target or evaluation reference. LoRA updates are trained with cross-entropy on assistant tokens, and deterministic task-specific formatting is applied after generation. No answer, task label, or reference timestamp is accessed during video indexing, retrieval, memory construction, or inference.

4.2 Label-Free Video Index

Let the 24 globally sampled frames be $\{I_k\}_{k=1}^K$, with timestamps t_k . In the reported configuration, consecutive grayscale frames provide a motion score

$$r_m(k) = \min \left(1, \frac{\text{mean} |I_k - I_{k-1}|}{64} \right), \quad (2)$$

and appearance novelty is the Bhattacharyya distance $r_n(k) \in [0, 1]$ between consecutive 32-bin grayscale histograms. The frame-level anomaly prior is not learned and does not use annotations:

$$r_a(k) = \frac{1}{2} \max\{r_m(k), r_n(k)\}. \quad (3)$$

Absent an external detector, the default index thresholds the grayscale difference image at 25, applies median filtering, and retains at most eight connected components whose area exceeds 0.2% of the image. Component area, normalized by 5% of image area and clipped to one, supplies detection confidence. Detections are greedily associated with the nearest active track when their normalized center distance is at most 0.12; tracks with fewer than two observations are discarded. For every pair of tracks with shared sampled frames, we compute their minimum center distance d_{jk} . Pairs with $d_{jk} \leq 0.18$ become interaction candidates with confidence

$$\rho_{jk} = \text{clip}\left(1 - \frac{d_{jk}}{0.18}, 0, 1\right). \quad (4)$$

The relation is labeled **crossing-path** when the tracks share at least two sampled frames and **approaching** otherwise. These labels are geometric hypotheses, not physical-causality estimates.

4.3 Saliency-Guided Temporal Retrieval

Candidate cores are two seconds long with a one-second stride. Each is expanded by two seconds on either side (and clipped to the video boundary) before its frames and features are collected. For expanded window w , we average the frame-level motion, novelty, and anomaly-prior scores; interaction saliency $r_i(w)$ is the maximum confidence of any candidate relation overlapping the window. The executed score is

$$s(w) = 0.15\bar{r}_m(w) + 0.25r_i(w) + 0.10\bar{r}_n(w) + 0.15\bar{r}_a(w). \quad (5)$$

The implementation adds a question-derived scalar to every window, but because that scalar is constant within a record it cancels from the ranking; Eq. 5 therefore states the effective, query-independent score. The coefficients are fixed configuration values rather than validation-tuned estimates. Greedy temporal non-maximum suppression rejects candidates with temporal IoU at least 0.5 against an already selected window and retains at most three windows. Records retain the indexed frames within those windows; the multimodal collator orders them chronologically and passes the first four available frames to the model. This final truncation is part of both training and evaluation.

4.4 Agent-Centric Event Memory

The full event-memory record $\mathcal{M}_V = (\mathcal{A}, \mathcal{R}, z, \Delta, c)$ combines agents $\mathcal{A}$, relations $\mathcal{R}$, a heuristic anomaly state z , interval Δ , and confidence vector c . Agent

a_j contains its generic category, track interval, normalized trajectory, inferred motion state, and mean detection confidence. A relation

$$r_{jk} = (a_j, p, a_k, t_s, t_e, \rho, \mathcal{F}_{jk}) \quad (6)$$

contains a predicate, overlap interval, confidence, and supporting frames. Relations sorted by start time form a candidate-event timeline. The highest confidence relation r^* additionally defines diagnostic metadata

$$z = \mathbb{I}[\rho^* \geq 0.65], \quad \Delta = \begin{cases} [t_s^*, t_e^*], & z = 1, \\ \emptyset, & z = 0. \end{cases} \quad (7)$$

The threshold is fixed, and both quantities depend only on the geometric relation miner. Crucially, z , Δ , and the heuristic cause field are stored for auditing but are *not* serialized into the model prompt. The model-visible memory $\widetilde{\mathcal{M}}_V$ contains only a generic scene statement, the relation timeline, track visibility intervals and confidences, and candidate relations with confidences. It closes with an instruction that the extracted evidence is uncertain and requires visual verification.

The same $\widetilde{\mathcal{M}}_V$ is reused for every $q_i \in \mathcal{Q}_V$; only the question and output schema vary. Unlike evidence-backed VideoQA, the current representation does not provide pixel-level evidence masks, and unlike learned graph-VideoQA architectures, its graph is neither end-to-end trained nor used for message passing. Its role is a lightweight, inspectable cross-question context interface.

4.5 Training Objective

For answer tokens $y_{1:L}$, answer-only teacher forcing masks system and user tokens and minimizes

$$\mathcal{L}_{\text{ans}} = - \sum_{\ell=1}^L \log p_{\theta}(y_{\ell} \mid y_{<\ell}, q, \widetilde{\mathcal{M}}_V, \mathcal{E}_i). \quad (8)$$

This answer-token loss is the complete objective used for the reported checkpoint; no grounding, consistency, or intervention loss is optimized.

5 Experimental Protocol

5.1 Data and Splitting

The official TAR release contains 44,040 annotations over 3,670 videos from eight public sources [1]. We ingest all released task files, normalize video identifiers, and group annotations by video before splitting. The parsed annotation inventory contained 3,669 TAR video identifiers and 590 auxiliary PSI-VQA references without corresponding media. Those unavailable out-of-domain references are excluded from the in-domain experiment. Media verification further

Table 1: Dataset reconciliation and frozen local experiment. PSI-VQA entries are excluded from the in-domain TAR evaluation; two TAR videos fail decoding.

Stage	Videos Records	
Official TAR release	3,670	44,040
Decode-valid frozen corpus	3,667	44,016
Training split	–	39,720
Validation split	–	4,296

Table 2: Number of validation records per task. Each one-per-video task has 358 records; BCQ and BCQ-O contain two questions per video.

Task	BCQ	MCQ	BCQ-O	MCQ-O	OQA	Cause	Scene	Temp.	Summ.	TLoc
Records	716	358	716	358	358	358	358	358	358	358

excludes two TAR videos that fail deterministic decoding; we report this reconciliation explicitly rather than silently imputing missing media. The frozen post-filter corpus therefore contains 3,667 usable TAR videos and 44,016 SFT records: 39,720 for training and 4,296 for validation. The seed-42 split is performed by normalized `video-id`, producing 3,309 training and 358 validation videos. The frozen manifests reconcile all 3,667 videos and 44,016 records, verify zero identifier overlap, and confirm that all local validation answers are present. The retained identifiers reconcile all reported videos and records.

5.2 Evaluation Metrics

The overall TAR score is computed as the unweighted mean of binary-question accuracy, multiple-choice accuracy, and BERTScore F1 across the seven open-ended reasoning tasks [1]. BERTScore evaluates semantic similarity between generated and reference answers using contextualized token representations [10]. Temporal localization is assessed separately using mean temporal intersection over union (m-tIoU) and is excluded from the aggregate TAR score. Because questions from the same video are dependent, uncertainty is estimated by re-sampling the 358 validation videos with replacement. We use 2,000 bootstrap replicates and percentile 95% intervals; paired replicates are used for adapted-minus-frozen categorical differences.

5.3 Baselines

The primary control is frozen Qwen3-VL-8B evaluated with the same records, four-frame limit, formatter, and decoding settings. This isolates LoRA adaptation under a matched evidence protocol. We also report a non-neural dataset-prior control: always predicting the most frequent validation label gives 50.1%

Table 3: Seed-42 local validation results on 4,296 post-filter records. BERTScore values are rescaled English RoBERTa-large F1. BCQ and MCQ denote binary and multiple-choice accuracy; the “-O” columns are their open-ended counterparts. OQA, Cause, Scene, Temp., and Summ. denote open question answering, causal linkage, scene description, temporal description, and video summarization, respectively. TAR is the unweighted mean over the nine scored tasks and excludes temporal localization.

Method	BCQ	MCQ	BCQ-O	MCQ-O	OQA	Cause	Scene	Temp.	Summ.	TAR	tIoU
Dataset-prior control	.501	.285	—	—	—	—	—	—	—	—	—
Frozen Qwen3-VL	.466	.617	.275	.490	.227	.153	.227	.083	.090	.292	.197
TRACE-TAR	.944	.883	.624	.672	.396	.424	.377	.348	.361	.559	.279

BCQ accuracy (“No,” 359/716) and 28.5% MCQ accuracy (option C, 102/358). This control diagnoses label imbalance but is not included in the TAR mean.

5.4 Implementation Details

We instantiate TRACE-TAR using Qwen3-VL-8B-Instruct and perform model training in `bf16`. Parameter-efficient adaptation is implemented with LoRA on the query, key, value, and output attention projections, as well as the gate, up, and down projections of the feed-forward layers. We use a LoRA rank of 64, a scaling factor of 128, and a dropout rate of 0.05.

The model is optimized with AdamW using a learning rate of 1×10^{-5} , a weight decay of 0.01, and a warm-up ratio of 0.03. Training is conducted for two epoch with gradient checkpointing on six NVIDIA A100 80 GB GPUs. Each GPU processes one multimodal record per step, and gradients are accumulated over four steps, yielding an effective batch size of 24. All experiments use seed 42.

During evaluation, generation is performed greedily with a maximum of 128 new tokens. Validation uses a per-device batch size of one to accommodate the memory requirements of long multimodal sequences. The adapted and frozen-model baselines are evaluated using the same retrieved frames, serialized event memory, prompting strategy, decoding configuration, and deterministic task-specific formatter, ensuring that their comparison isolates the effect of supervised LoRA adaptation.

6 Results and Diagnostic Analysis

Matched frozen-model comparison. Under identical evidence and decoding, LoRA adaptation raises the nine-task mean from 0.2922 to 0.5587, an absolute difference of 0.2665. All nine scored task metrics and diagnostic temporal mIoU improve. Both systems receive the same saliency-selected frames and serialized memory, so the comparison measures the effect of supervised adaptation under matched evidence. The BCQ and MCQ estimates have video-bootstrap 95% intervals of [0.933, 0.965] and [0.863, 0.927], respectively; the paired adapted-minus-frozen differences are [0.446, 0.517] for BCQ and [0.224, 0.335] for MCQ.

Table 4: Diagnostic failure analysis on the saved validation predictions. These measures are not official leaderboard metrics.

Failure	Evidence	Implication
Cross-task contradiction	BCQ versus BCQ-O agreement: 53.9% across 716 pairs.	Shared context alone does not enforce compatible answers.
Cause–summary divergence	Token IoU: 0.148 across 358 pairs.	The two outputs often emphasize different participants or event details.
Temporal error	Mean temporal IoU: 0.279; timestamp agreement within 2s: 0.6%.	Event recognition is substantially stronger than temporal localization.
Entity substitution	A motorcycle is replaced by a black car in one audited example.	An incorrect identity can propagate across several task outputs.

These intervals treat the video, rather than each correlated question, as the resampling unit.

Cross-task prediction diagnostic. On the saved TRACE-TAR predictions, categorical agreement is 53.9% across 716 BCQ/open-BCQ pairs and 95.8% across 358 MCQ/open-MCQ pairs. Token IoU between summary and causal answers is 0.148 over 358 pairs, while only 0.6% of 358 localization/temporal-description pairs contain timestamps within two seconds. These deterministic, reference-free diagnostics are not official TAR metrics; they show that high task accuracy does not by itself ensure a coherent shared event description.

Structured failure analysis. Table 4 organizes observed failures by the capability they test. Categorical accuracy is strongest, but BCQ/open-BCQ agreement is only 53.9%, so correct individual predictions need not form a coherent video-level account. Open-ended BERTScores range from 0.348 to 0.672, and temporal mIoU is 0.279, identifying description and localization as the principal bottlenecks.

In the entity-substitution case, the model correctly identifies a blue-underglow vehicle and selects the correct multiple-choice option, but uses the wrong road user for the 7–11 s temporal and causal questions, illustrating how an identity error propagates across task-specific answers.

7 Limitations, Ethics, and Broader Impact

Event memory quality is bounded by frame sampling, detection, and tracking. The relation miner uses geometric proximity and motion cues; it cannot establish

physical causality and its outputs are explicitly uncertain. Generated answers can contain fluent but unsupported reasoning, and BERTScore may reward semantic similarity despite incorrect agents or times.

Transportation surveillance data can expose identities, locations, and sensitive behavior. Our pipeline retains only benchmark-authorized data paths and derived model inputs, and does not introduce identity recognition. A model trained on geographically or environmentally imbalanced footage may fail under different weather, camera, or road-user distributions. Predictions should support human analysis, not autonomously assign fault or trigger enforcement.

8 Conclusion

We presented a transportation-anomaly reasoning pipeline that reuses label-free temporal evidence and uncertain track-relation context across the questions associated with a video. On a video-disjoint local split, LoRA adaptation raises the nine-task mean from 0.2922 to 0.5587 under a matched input protocol. Error analysis identifies temporal localization, open-ended description, and cross-task identity consistency as the primary challenges.

References

1. AI City Challenge: 2026 track 3: Anomalous events in transportation. <https://www.aicitychallenge.org/2026-track3/> (2026), accessed 23 July 2026
2. Bai, S., Cai, Y., Chen, R., et al.: Qwen3-vl technical report. arXiv preprint arXiv:2511.21631 (2025)
3. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: Lora: Low-rank adaptation of large language models. In: International Conference on Learning Representations (2022)
4. Huang, C., Wang, B., Wen, J., Liu, C., Wang, W., Shen, L., Cao, X.: Vad-r1: Towards video anomaly reasoning via perception-to-cognition chain-of-thought. ArXiv **abs/2505.19877** (2025), <https://api.semanticscholar.org/CorpusID:278910552>
5. Huang, L., Chen, W., Chen, Z., Liu, Y., Lin, L.: Eg-vqa: Benchmarking verifiable video question answering with grounded temporal evidence. arXiv preprint arXiv:2606.24797 (2026)
6. Landi, F., Snoek, C.G.M., Cucchiara, R.: Anomaly locality in video surveillance. vol. abs/1901.10364 (2019), <https://api.semanticscholar.org/CorpusID:59336279>
7. Wang, S., Zhou, H., Wang, Z., Xu, R., Xiong, C., Savarese, S., Sun, C., Niebles, J.C.: Evidence-backed video question answering. arXiv preprint arXiv:2607.11862 (2026)
8. Xiao, J., Yao, A., Liu, Z., Li, Y., Ji, W., Chua, T.S.: Video as conditional graph hierarchy for multi-granular question answering. In: AAAI Conference on Artificial Intelligence (2022)
9. Zhang, H., Sun, A., Jing, W., Zhou, J.T.: Temporal sentence grounding in videos: A survey and future directions. IEEE Transactions on Pattern Analysis and Machine Intelligence (2023)

10. Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: Bertscore: Evaluating text generation with bert. In: International Conference on Learning Representations (2020)